

Multimodal Fusion with Generative AI-Enhanced Textual Features for Multi-Horizon Crude Oil Price Forecasting

Guan-Yan Yang^a, Wei-Yang Chiu^b, Farn Wang^{a,*} and Kuo-Hui Yeh^{b,c,*}

^aDepartment of Electrical Engineering, National Taiwan University, No. 1, Sec. 4, Roosevelt Rd., Taipei, 106319, Taiwan (R.O.C.)

^bInstitute of Artificial Intelligence Innovation, National Yang Ming Chiao Tung University, No. 1001, Daxue Road, East District, Hsinchu City, 300093, Taiwan (R.O.C.)

^cDepartment of Information Management, National Dong Hwa University, No. 1, Sec. 2, Da Hsueh Road, Shoufeng, Hualien, 974301, Taiwan (R.O.C.)

ARTICLE INFO

Keywords:

Crude Oil Price Forecasting
Multimodal Fusion
Deep Learning
Financial News Sentiment
Long Short-Term Memory (LSTM)
Generative AI

ABSTRACT

Crude oil price forecasting is a critical yet challenging task for financial risk management and energy procurement, given the market's inherent volatility and complex dependencies on economic and geopolitical factors. Traditional forecasting models often struggle to integrate the diverse data modalities that drive price dynamics. This paper proposes a novel decomposition-based multimodal deep learning framework that enhances crude oil price prediction by systematically fusing structured numerical data with enriched, unstructured textual features. The framework integrates over 9,000 macroeconomic and commodity-related indicators with financial news features. To overcome the limitations of sparse news headlines, we employ a generative AI approach using GPT-4.1 to produce context-aware summaries, which are then encoded into dense embeddings via FinBERT. The West Texas Intermediate (WTI) price series is decomposed into trend, seasonal, and residual components, with each component modeled by a dedicated Long Short-Term Memory (LSTM) network using tailored inputs. We systematically implement and evaluate three distinct fusion strategies—early, intermediate, and late—to identify the optimal architecture for integrating heterogeneous data. A custom hybrid loss function is introduced, which balances prediction error minimization with directional accuracy to improve trend sensitivity. Experimental results show that the proposed framework, particularly the early fusion strategy, significantly outperforms unimodal and limited-modality baselines in forecasting WTI prices up to eight weeks ahead. The early fusion model demonstrates superior performance across MSE, MAPE, and SMAPE metrics. This study provides a robust, interpretable, and adaptive decision support system for navigating the complexities of the crude oil market.

1. Introduction

The crude oil market stands as a cornerstone of the global economy, where price fluctuations exert profound impacts on inflation, industrial production, and national energy policies [1]. For stakeholders in the energy sector, from governments to corporations, accurate price forecasting is not merely an academic exercise but a critical component of strategic planning, risk management, and procurement optimization. However, forecasting crude oil prices is notoriously difficult due to their nonlinear, non-stationary nature, which is driven by a complex interplay of supply-demand dynamics, macroeconomic shifts, geopolitical events, and financial speculation [2]. Events such as the COVID-19 pandemic and geopolitical conflicts have starkly demonstrated the market's susceptibility to extreme volatility, underscoring the urgent need for robust forecasting models that can provide reliable guidance for strategic decision-making [3].

Traditional econometric models like ARIMA and GARCH, while useful for capturing linear trends and volatility clustering, are often insufficient for modeling the complex, nonlinear dynamics of the oil market [4, 5]. A primary limitation is their typically univariate nature, which prevents them from incorporating crucial external factors. While multivariate extensions like Vector Autoregression (VAR) and ARIMAX can integrate macroeconomic indicators (such as interest rates, stock indices), they are constrained by linearity assumptions and can be prone to overparameterization, limiting their robustness in capturing the market's non-stationary behavior [6].

To address these limitations, machine learning and deep learning models have gained prominence. Models such as Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Temporal Convolutional Networks (TCN) have demonstrated superior performance by capturing complex temporal dependencies from multivariate numerical data [7, 8]. However, a significant portion of these studies neglect the rich, qualitative information embedded in unstructured textual data, such as financial news. News sentiment and event-driven narratives often contain signals that precede significant price shifts, yet numerical data alone cannot capture them [9].

While some recent works have started to incorporate textual data, they often face significant challenges. A common approach is to reduce news content to a single sentiment

*Corresponding author

✉ f11921091@ntu.edu.tw (G. Yang); weich@nycu.edu.tw (W. Chiu); farn@ntu.edu.tw (F. Wang); khyeh@nycu.edu.tw (K. Yeh)

🌐 <https://ianyang66.github.io/> (G. Yang); <https://waynechiu.cc/> (W. Chiu); <https://khyeh.web.nycu.edu.tw/> (K. Yeh)

ORCID(s): 0009-0002-2539-9057 (G. Yang); 0000-0002-8917-4087 (W. Chiu); 0000-0002-0425-6500 (F. Wang); 0000-0003-0598-761X (K. Yeh)

¹Received 2018; Received in revised form 2018; Accepted 2018.
Available online 2018

score, which fails to capture the full contextual meaning of an event [10, 11]. Furthermore, many studies rely on short news headlines, which are difficult to model effectively [12]. The advent of Large Language Models (LLMs) offers a transformative solution. Generative AI, through sophisticated prompt engineering, can convert sparse headlines and lengthy articles into structured, context-rich summaries, significantly enhancing the quality of textual features for forecasting models [13].

Despite these advances, a critical research gap remains in the systematic fusion of these heterogeneous data modalities. Many multimodal forecasting models process each data type in isolation and combine them only at a late stage, overlooking the deep, interactive relationships between them [14]. The optimal strategy for integrating numerical time-series with high-dimensional textual embeddings is still an open question in the energy forecasting domain.

This paper addresses these challenges by proposing a comprehensive, decomposition-based multimodal fusion framework for multi-horizon crude oil price forecasting. Our work makes the following key contributions:

1. **Generative AI-Enhanced Textual Features:** We pioneer the use of prompt engineering with GPT-4.1 to transform raw news articles into high-quality, context-aware summaries. These summaries are then encoded using FinBERT, a domain-specific language model, to generate rich sentence embeddings that capture nuanced market sentiment and event context, moving beyond simple polarity scores.
2. **Decomposition-based Multimodal Architecture:** We employ Seasonal-Trend decomposition using Loess (STL) to break down the WTI price series into interpretable trend, seasonal, and residual components. Each component is modeled by a dedicated LSTM network fed with tailored inputs—macroeconomic and textual features for the trend, historical lags for seasonality, and event-driven textual features for the residual—enhancing both accuracy and interpretability.
3. **Systematic Fusion Strategy Evaluation:** We implement and rigorously compare three distinct multimodal fusion architectures: early, intermediate, and late fusion. This systematic evaluation identifies the most effective strategy for integrating numerical and textual data for crude oil forecasting, a comparison largely absent in prior literature. Our results demonstrate that early fusion consistently provides the most accurate and stable forecasts.
4. **Actionable Multi-Horizon Forecasting:** The proposed model is designed to provide forecasts up to eight weeks in advance, directly addressing the practical needs of procurement managers and financial analysts who operate on longer planning cycles. A custom hybrid loss function that penalizes both prediction error and directional inaccuracy further ensures the model's practical utility.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on crude oil price forecasting. Section 3 details the proposed multimodal fusion framework. Section 4 presents the experimental setup and discusses the results. Finally, Section 5 concludes the paper and outlines directions for future research.

2. Related Work

Research in crude oil price forecasting has evolved significantly, progressing from purely statistical models to sophisticated multimodal deep learning frameworks. This evolution can be categorized into four key areas: (1) traditional time-series and unimodal deep learning models, (2) signal decomposition techniques, (3) the integration of textual data, and (4) multimodal fusion strategies. Table 1 offers a comparative summary of seminal works and situates our study within this evolving landscape.

2.1. Time-Series, Structural, and Unimodal DL Models

Early forecasting efforts were dominated by statistical time-series models like the Autoregressive Integrated Moving Average (ARIMA) [15] and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) [16, 5], which model future prices based on their own past values and volatility patterns. While foundational, their typical inability to incorporate external economic information is a major drawback [17]. Structural models, such as Vector Autoregression (VAR), attempted to overcome this by including quantitative economic factors [6, 18], but they are often constrained by linearity assumptions and can struggle with the "curse of dimensionality" when many predictors are used.

The rise of machine learning introduced more flexible approaches. Hybrid models combining neural networks with optimization methods like genetic algorithms were among the first to demonstrate improved performance [19]. Subsequently, deep learning brought powerful unimodal models like Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), which excel at capturing complex, non-linear temporal dependencies from price history and other numerical inputs [20, 21, 22]. Recent studies have continued to refine this numerical-only approach; for example, Zhang et al. in 2024 [23] developed a sophisticated forecast combination strategy that uses attention-weighted Principal Component Analysis (PCA) to identify the most salient numerical predictors. However, these models remain fundamentally limited by relying solely on quantitative data, thereby ignoring the rich, event-driven information embedded in textual sources.

2.2. Signal Decomposition for Non-Stationarity

To address the highly non-stationary and complex nature of oil prices, signal decomposition techniques, often framed within a "divide and conquer" methodology, have become widespread. As pioneered by Yu et al. in 2015 [24], this approach involves breaking down a time series into

several simpler sub-components, which are then modeled individually before being aggregated. Common methods include the Wavelet Transform (WT) [25, 26] and Empirical Mode Decomposition (EMD) with its variants like Ensemble EMD (EEMD) and Complete EEMD with Adaptive Noise (CEEMDAN) [27, 28].

More recently, a trend has emerged towards using secondary or dual decomposition to achieve a finer-grained separation of signal components. For instance, Gong et al. in 2024 [29] and Zhu et al. in 2025 [30] both propose frameworks where the price series is first decomposed by CEEMDAN, and then the highest-frequency (noisiest) component is further decomposed using Variational Mode Decomposition (VMD). Similarly, Li et al. in 2021 [31] applied VMD to analyze the impact of sentiment on oil price volatility. While these advanced techniques are powerful, the resulting components can lack clear economic interpretability. Seasonal-Trend decomposition using Loess (STL) offers a compelling alternative by separating a series into distinct and interpretable trend, seasonal, and residual components [32], a structure that allows for a more targeted assignment of predictive features, which we adopt in this study.

2.3. Integrating Textual Data and Sentiment

Recognizing that market movements are often driven by news and public perception [2], researchers began integrating textual data. Initial approaches focused on sentiment analysis, converting news into simple polarity scores. Li et al. in 2019 [9] represented an early effort using deep learning (CNNs) for text-based forecasting, though their features were based on shallow TextBlob polarity. A significant advancement came with the use of online search engine data, as demonstrated by Tang et al. in 2020 [33], who integrated multi-factor search data with multi-scale analysis.

More advanced methods have emerged using topic modeling [10, 34] or deep learning-based embeddings. The work by Fang et al. in 2023 [11] marked a significant step forward by using FinBERT [35], a domain-specific language model, to create nuanced vector representations that capture financial context far better than generic models. However, these models still often rely on short, potentially uninformative news headlines, which are known to be challenging for topic modeling [12]. Our work advances this area by using generative AI (GPT-4.1) to create high-quality, structured summaries from full news articles. This approach provides a richer and more condensed input for FinBERT encoding, moving beyond simple sentiment scores to capture deeper event context, a methodology supported by recent work on using LLMs for structured information extraction [13, 36].

2.4. Multimodal Fusion Strategies

Multimodal learning aims to build models that can jointly reason over and learn from heterogeneous data sources. In finance, this typically involves combining numerical time series with textual data. Fusion strategies are generally categorized as early, intermediate, or late [37, 38].

- **Early fusion** concatenates features at the input level, allowing a single model to learn cross-modal interactions from the start. This approach has been used in crude oil forecasting, as seen in the work of Bai et al. in 2022 [10] and Fang et al. in 2023 [11].
- **Intermediate fusion** processes each modality in a separate stream before merging their learned latent representations at a middle layer. While this is a common approach in financial forecasting for assets like stocks [14, 39] and exchange rates [40], its application in the crude oil domain is less explored. The work by Akil & Khedr in 2024 [41] employs this strategy but focuses solely on fusing different types of numerical data, excluding text.
- **Late fusion** trains independent models for each modality and combines their final predictions, often through averaging or a simple linear layer [42]. This is conceptually similar to the ensemble or combination methods used in many numerical-only frameworks [29, 23, 30].

While intermediate fusion is a common choice in general financial forecasting, very few studies in the crude oil domain have systematically compared these strategies to identify the optimal approach. Our research directly addresses this gap by implementing and rigorously evaluating all three fusion architectures within a unified, decomposition-based framework that leverages generative AI for superior feature engineering.

3. Methodology

The proposed research framework, depicted in Figure 1, is a multi-stage process designed to forecast WTI crude oil prices by fusing numerical and textual data. The five stages are: (1) Data Collection, (2) Text Feature Generation, (3) Data Preprocessing, (4) Multimodal Fusion, and (5) Multi-Step Forecasting.

3.1. Data Collection and Feature Generation

3.1.1. Data Sources

We collected data from January 2015 to December 2019, a period preceding the extreme volatility of the COVID-19 pandemic, to build a model based on more typical market dynamics. Three categories of data were gathered:

- **Crude Oil Prices:** Daily WTI crude oil spot prices were used as the target variable.
- **Numerical Indicators:** A comprehensive dataset of 9,390 features was compiled from sources such as the U.S. Energy Information Administration (EIA), OECD-STAT, and the Federal Reserve Economic Data (FRED). These include energy transaction data, production/reserve levels, and a wide array of macroeconomic indicators (such as CPI, PMI, interest rates).

Table 1

Comparison of recent crude oil price forecasting models.

Reference	Methodology	Textual Features	Numerical Features	Decomposition	Fusion Strategy	Key Contribution / Limitation
Yu et al. [24] (2015)	Decomposition-Ensemble	No	Yes	Yes (Custom Reconstruction)	Ensemble	Data-characteristic-driven reconstruction of decomposed modes to improve accuracy and efficiency.
Li et al. [9] (2019)	SVR, RF, CNN	TextBlob polarity, CNN topic scores	Yes	No	None (Sequential)	Early use of deep learning for text-based forecasting. / Topic scores are shallow.
Tang et al. [33] (2020)	Multi-scale Forecasting	Yes (Search engine data)	Yes	Yes (MEMD)	Ensemble	Integrates multi-factor search data with multi-scale analysis. / Search data may lack specific event context.
Abdollahi [28] (2020)	Hybrid Model	No	Yes	Yes (CEEMD)	Ensemble/Hybrid	Sophisticated hybrid of CEEMD, SVM, PSO, and GARCH to capture non-linearity and volatility. / High model complexity.
Li et al. [31] (2021)	BiLSTM	Cumulative sentiment score	No	Yes (VMD)	None (Unimodal)	Models asymmetric effects of sentiment on volatility. / Sentiment index is manually constructed.
Bai et al. [10] (2022)	AdaBoost.RT	SeaNMF topic vector, dynamic sentiment	No	No	Early Fusion	Advanced textual indicators for short headlines. / Excludes numerical data.
Fang et al. [11] (2023)	BiGRU-Attention	FinBERT sentiment score	Yes	Yes (VMD)	Early Fusion	Integrates sentiment with price and economic data. / Decomposes only the price series.
Akil & Khedr [41] (2024)	LSTM, TCN	No	Yes	No	Intermediate Fusion	Multimodal deep learning with economic indicators. / Excludes textual data, short forecast horizon.
Gong et al. [29] (2024)	CEEMDAN-VMD + BiGRU-Att-CNN	No	Yes	Yes (CEEMDAN-VMD)	Ensemble	Advanced secondary decomposition framework. / Excludes textual data.
Zhang et al. [23] (2024)	Attention-PCA + ABC Fusion	No	Yes	No (PCA is feature extraction)	Ensemble/Combination	Sophisticated forecast combination using attention-weighted features. / Does not compare fusion architectures.
Zhu et al. [30] (2025)	CEEMDAN-VMD + CNN-BiLSTM	No	Yes	Yes (CEEMDAN-VMD)	Ensemble	Dual decomposition with hybrid deep learning. / Excludes textual data.
This Study	LSTM	GPT-4.1 summaries + FinBERT embeddings	Yes (9000+)	STL	Early, Intermediate, Late Fusion	Generative AI for feature enhancement; systematic fusion comparison; interpretable decomposition.

- **Textual Data:** We collected over 9,000 financial news articles from the "Commodities" section of Investing.com for the period of our study. Rather than relying on shallow metrics, we leveraged a generative AI approach to systematically distill this corpus. For each article, we used the GPT-4.1 API with carefully engineered prompts to generate a concise, context-aware summary. This process transforms lengthy articles into structured, information-dense text that highlights the core market-moving information. This step is crucial, as it elevates the textual data from a simple sentiment signal to a rich contextual feature, making it probably a powerful complement to the numerical data.

3.1.2. Generative AI for Text Feature Enhancement

To overcome the limitations of sparse news headlines, we utilized the GPT-4.1 API to generate enriched textual features from the full content of each news article. By prompting the model to act as an "economist and journalist," we generated three features for each article: a concise headline, five salient keywords, and a one-line summary. This process transforms unstructured text into semantically rich, model-ready inputs.

3.2. Data Preprocessing

3.2.1. Numerical Data Preprocessing

The numerical data underwent several preprocessing steps. Missing values, primarily from weekends and holidays, were handled using linear interpolation. All features were transformed to a daily frequency. We then applied Seasonal-Trend decomposition using Loess (STL) to the

WTI price series, breaking it into trend (T_t), seasonal (S_t), and residual (R_t) components:

$$P_t = T_t + S_t + R_t \quad (1)$$

The Augmented Dickey-Fuller (ADF) test was used to check for stationarity. The non-stationary trend component was made stationary by applying first-order differencing. To manage the high dimensionality of the numerical indicators, we employed the Granger causality test to select only the features that have a statistically significant predictive relationship with the target components. Finally, all selected numerical features were standardized using StandardScaler.

3.2.2. Textual Data Preprocessing

The textual data were cleaned by converting to lower-case, removing punctuation, and filtering stop words. We then used **CrudeBERT** [43], a sentiment analysis model fine-tuned for the oil market, to classify the sentiment of each text. To create a representative weekly textual feature, we first determined the dominant daily sentiment (positive or negative) by majority vote across multiple news articles. We then selected the single most representative article for that day based on the highest CrudeBERT softmax probability. The GPT-generated summaries from these seven daily articles were concatenated to form a single weekly text input.

This weekly summary was then converted into a dense vector representation using **FinBERT** [35], a BERT model pre-trained on financial text. We applied masked mean pooling to the token-level embeddings to generate a single 768-dimensional sentence embedding for each week, capturing the deep contextual meaning of the news.

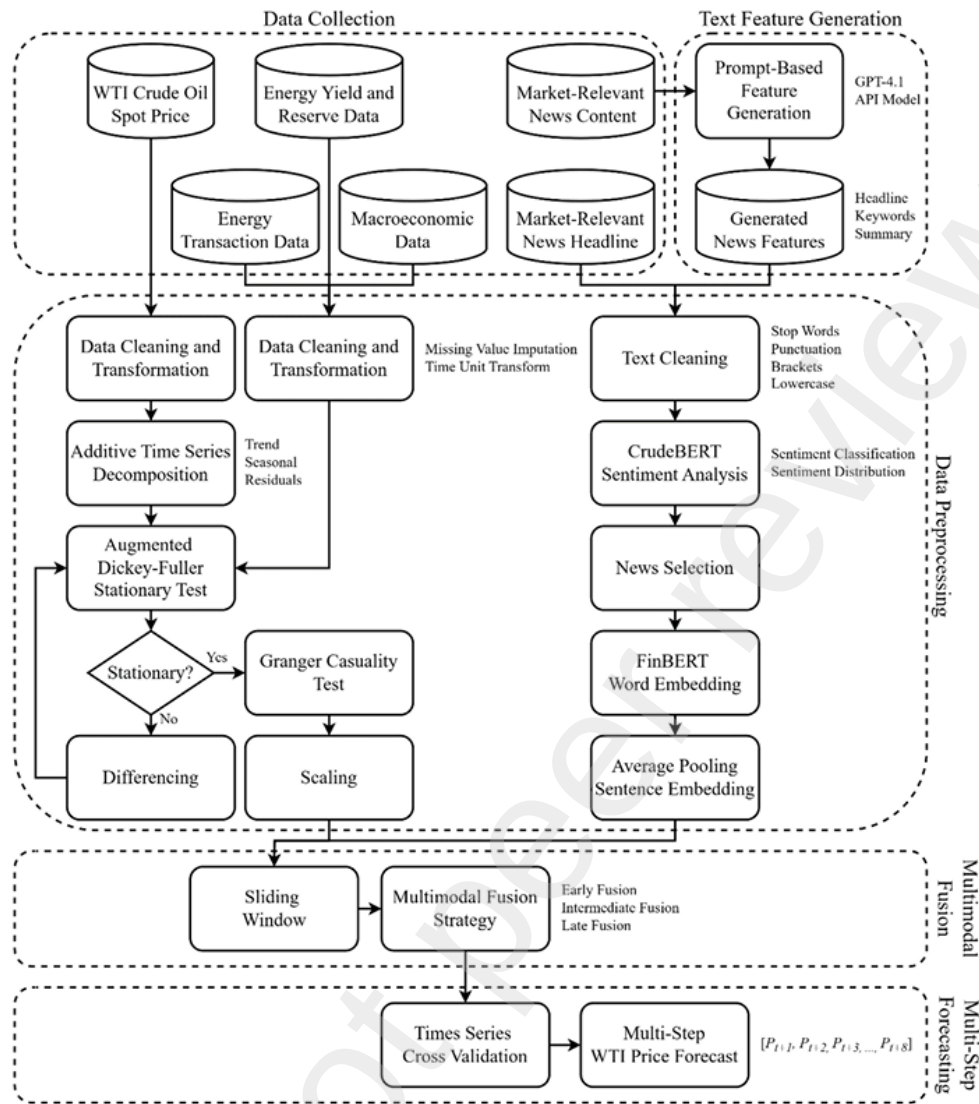


Figure 1: Detailed framework of the multimodal oil forecasting model.

3.3. Multimodal Fusion and Forecasting

We developed a decomposition-based forecasting architecture where each price component (trend, seasonal, residual) is predicted by a separate LSTM-based model. The inputs to these models are tailored to the characteristics of each component:

- **Trend Model:** Fuses lagged trend differences, selected macroeconomic features, and textual embeddings.
- **Seasonal Model:** Uses only lagged seasonal values, as seasonality is primarily driven by its own history.
- **Residual Model:** Fuses lagged residual values and textual embeddings to capture event-driven shocks.

We systematically evaluated three fusion strategies for the trend and residual models:

1. **Early Fusion:** All numerical and textual features are concatenated at the input layer and fed into a single LSTM network (Figure 2).
2. **Intermediate Fusion:** Each modality is processed by a separate LSTM branch, and their latent representations are concatenated and passed through a dense layer for prediction (Figure 3).
3. **Late Fusion:** Separate LSTM models are trained for each modality. Their individual predictions are then concatenated and fed into a final dense layer to produce the combined forecast (Figure 4).

The model uses a sliding window approach, taking the past eight weeks of data to predict the next eight weeks. The final price forecast is obtained by summing the predictions of the three component models.

3.4. Hybrid Loss Function

To improve both predictive accuracy and directional correctness, we designed a hybrid loss function. The total

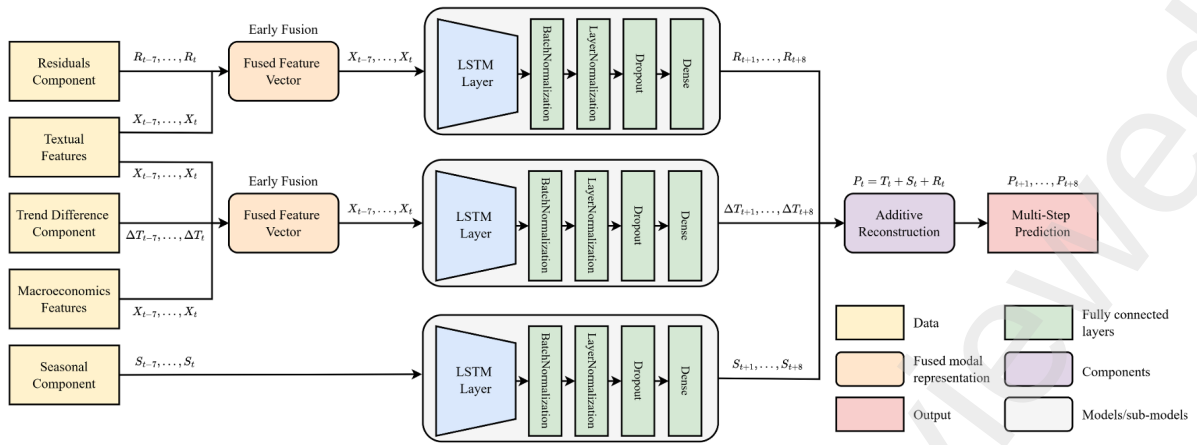


Figure 2: Early fusion architecture for multimodal time-series forecasting.

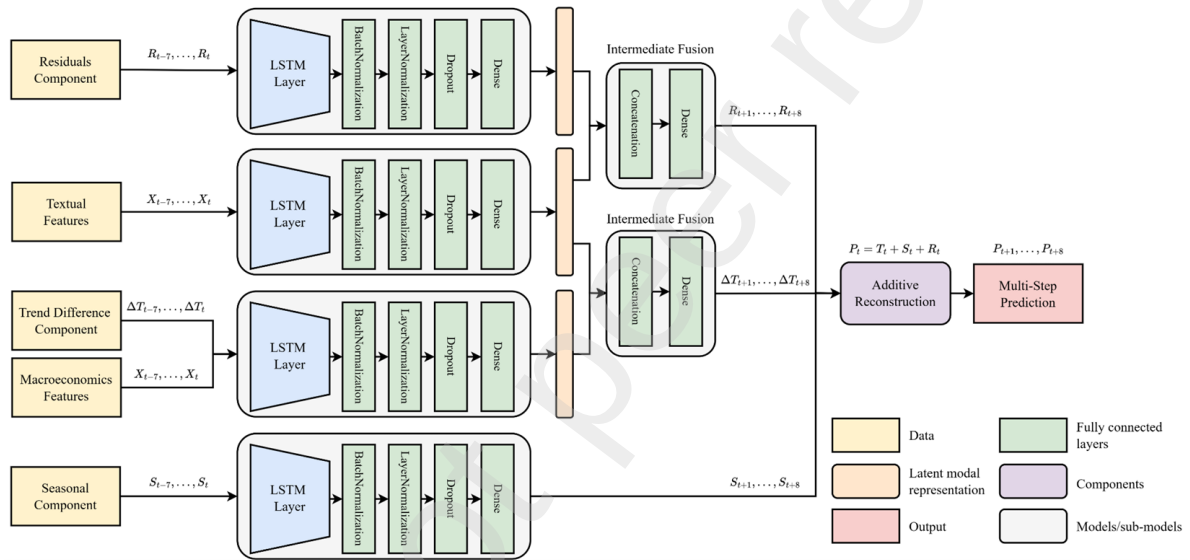


Figure 3: Intermediate fusion architecture for multimodal time-series forecasting.

loss is a weighted sum of a standard error loss (MSE or Huber) and a directional penalty term.

$$L_{\text{hybrid}} = \lambda_1 \times L_{\text{base}} + \lambda_2 \times L_{\text{dir}} \quad (2)$$

For the noisy residual component, we use Mean Squared Error (MSE) as the base loss (L_{base}). For the trend and seasonal components, we use the more robust Huber loss. The directional penalty, L_{dir} , is applied only to the trend and seasonal components and penalizes the model when the predicted direction of change mismatches the actual direction, scaled by the magnitude of the error. This encourages the model to learn not just the price level but also its trajectory.

3.5. Evaluation Metrics

Model performance was evaluated using Mean Squared Error (MSE), Mean Absolute Percentage Error (MAPE), and Symmetric Mean Absolute Percentage Error (SMAPE). We also measured **Trend Accuracy**, defined as the percentage of predictions where the model correctly identifies the direction

of price movement (rise, fall, or flat) relative to a $\pm 1\%$ threshold.

4. Experiments and Results

We conducted a series of experiments to evaluate the proposed framework. First, we compared the performance of the three fusion strategies against five baseline models: (1) WTI price only, (2) other numerical features only, (3) textual features only, (4) WTI price + textual features, and (5) WTI price + other numerical features. Second, we benchmarked our best-performing model against baseline methods from recent literature to demonstrate its competitiveness and contribution to the field.

4.1. Analysis of Selected Features and Decomposition

The Granger causality test identified 430 numerical features as significant predictors for the trend component. The

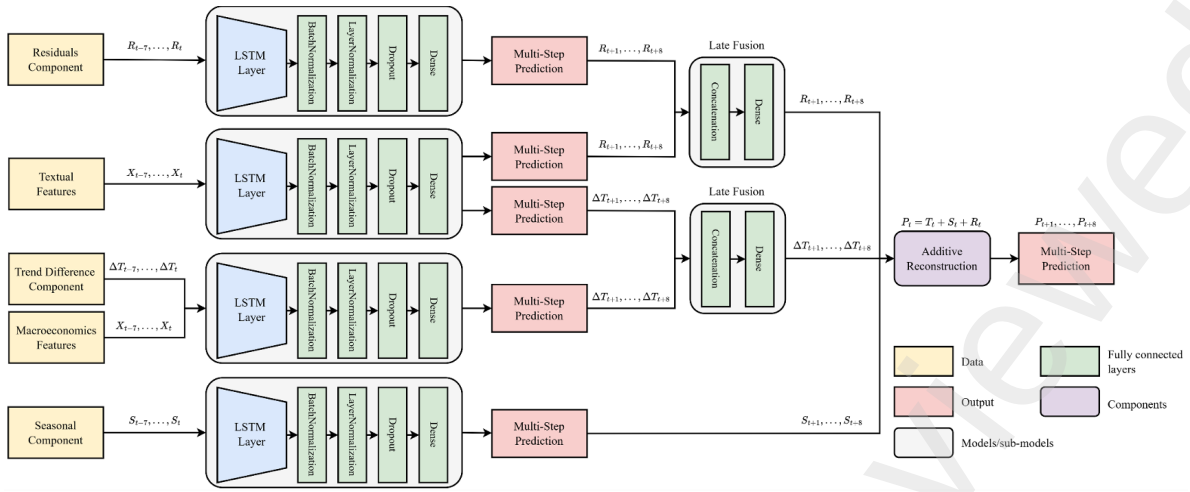


Figure 4: Late fusion architecture for multimodal time-series forecasting.

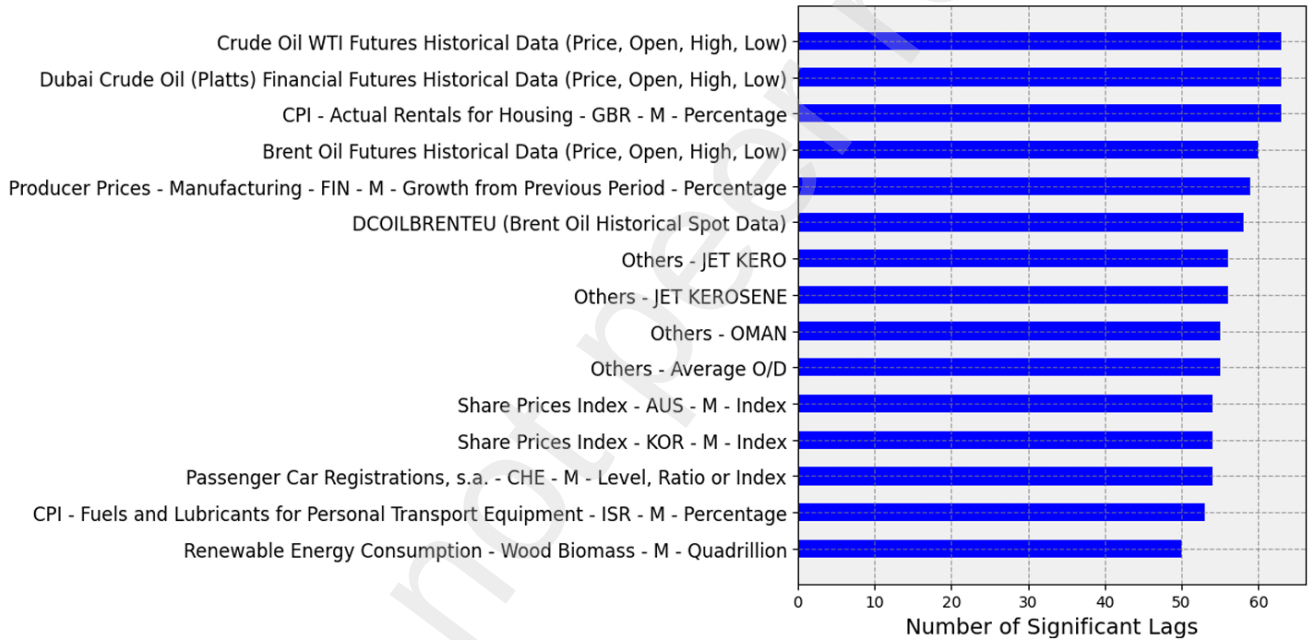


Figure 5: Top 15 most influential variables for oil price trend forecast based on Granger causality.

top 15 features, shown in Figure 5, are dominated by energy market indicators (such as WTI, Brent, Dubai futures), refined product prices, and key macroeconomic variables such as CPI and PPI. This confirms that the feature selection process identified theoretically sound and practically relevant drivers of oil prices. The STL decomposition successfully separated the WTI price into a clear long-term trend, a recurring 28-day seasonal pattern, and a high-frequency residual component, providing an interpretable basis for the component-wise modeling.

4.2. Comparison of Modalities and Fusion Strategies

Tables 2, 3, 4, and 5 summarize the forecasting performance for all models across the eight-week forecast horizon.

The results consistently demonstrate the superiority of multimodal fusion models over all baselines. The **Early Fusion** model emerges as the top performer in terms of error metrics (MSE, MAPE, and SMAPE) across all eight weeks. For instance, in Week 8, it achieves a MAPE of 5.78%, which is a 0.62% improvement over the strongest baseline (Crude Oil + Numerical Features). This indicates that integrating all data modalities at the input level allows the LSTM to learn the most effective joint representations and capture complex cross-modal dependencies.

The **Intermediate Fusion** model is the second-best performer. While its error metrics are slightly higher than early fusion, it exhibits the lowest standard deviation in the first three weeks, suggesting high stability for short-term forecasting. However, its performance variability increases

Table 2

MAPE-based performance comparison across modalities (%). Lower is better. Standard deviation in parentheses.

Metric (MAPE %)	Crude Oil Price	Other Num. Features	Textual Features	Crude Oil + Textual	Crude Oil + Num.	Early Fusion	Intermediate Fusion	Late Fusion
Week 1	1.93 (0.34)	1.72 (0.21)	2.01 (0.47)	1.65 (0.27)	1.62 (0.25)	1.54 (0.20)	1.58 (0.17)	1.63 (0.13)
Week 2	3.14 (0.63)	2.27 (0.38)	3.28 (0.79)	2.61 (0.35)	2.10 (0.52)	1.92 (0.25)	2.06 (0.12)	2.22 (0.27)
Week 3	4.69 (0.91)	3.09 (0.58)	4.90 (1.03)	4.12 (0.87)	2.98 (0.52)	2.63 (0.46)	2.76 (0.32)	2.97 (0.47)
Week 4	5.84 (1.29)	3.82 (0.62)	6.05 (1.43)	5.40 (1.07)	3.64 (0.70)	3.45 (0.65)	3.56 (0.66)	3.80 (0.74)
Week 5	7.11 (1.71)	4.73 (0.92)	7.22 (1.98)	6.66 (1.47)	4.54 (1.00)	4.51 (0.86)	4.55 (1.09)	4.88 (1.08)
Week 6	7.91 (2.00)	5.27 (1.17)	7.86 (2.42)	7.40 (1.88)	5.04 (1.28)	4.98 (1.03)	5.06 (1.59)	5.45 (1.51)
Week 7	8.50 (2.27)	5.70 (1.61)	8.22 (2.83)	7.88 (2.26)	5.67 (1.76)	5.35 (1.39)	5.45 (1.94)	5.92 (1.88)
Week 8	9.21 (2.71)	6.45 (2.11)	8.74 (3.38)	8.34 (2.72)	6.39 (2.20)	5.78 (1.91)	6.05 (2.49)	6.54 (2.39)

Table 3

MSE-based performance comparison across modalities. Lower is better. Standard deviation in parentheses.

Metric (MSE)	Crude Oil Price	Other Num. Features	Textual Features	Crude Oil + Textual	Crude Oil + Num.	Early Fusion	Intermediate Fusion	Late Fusion
Week 1	2.06 (0.56)	1.69 (0.39)	2.28 (0.80)	1.56 (0.44)	1.57 (0.37)	1.37 (0.26)	1.44 (0.15)	1.56 (0.20)
Week 2	5.30 (1.94)	2.98 (0.81)	5.91 (2.59)	3.86 (1.33)	2.58 (0.72)	2.26 (0.66)	2.58 (0.38)	2.92 (0.85)
Week 3	11.57 (4.61)	5.32 (1.52)	12.69 (5.60)	9.27 (3.83)	5.07 (1.60)	4.08 (1.33)	4.53 (0.78)	5.13 (1.62)
Week 4	18.49 (8.06)	8.22 (2.06)	19.99 (9.78)	16.34 (7.46)	7.71 (2.35)	6.81 (2.19)	7.21 (1.99)	8.24 (2.90)
Week 5	28.10 (12.75)	13.49 (4.76)	29.46 (16.21)	25.68 (13.30)	13.08 (5.51)	12.12 (4.15)	12.79 (5.24)	14.17 (5.95)
Week 6	35.51 (16.52)	17.84 (8.61)	36.36 (22.44)	33.03 (19.63)	17.05 (9.24)	15.68 (6.60)	17.35 (10.03)	19.42 (10.76)
Week 7	41.32 (19.91)	22.64 (14.27)	41.28 (28.27)	38.81 (24.94)	22.35 (14.78)	19.35 (10.64)	21.82 (14.58)	24.55 (16.21)
Week 8	48.27 (24.98)	29.53 (20.48)	47.56 (35.74)	44.62 (30.38)	28.30 (19.96)	23.59 (15.74)	27.65 (21.50)	30.74 (22.84)

in later weeks. The **Late Fusion** model performs the weakest among the fusion strategies, highlighting the importance of learning interactions between modalities during the training process rather than simply combining outputs.

Regarding Trend Accuracy, the models incorporating structured numerical features perform significantly better than those relying on textual data alone. This suggests that while textual features are crucial for reducing overall prediction error and capturing event-driven volatility, the fundamental direction of price trends is more strongly dictated by macroeconomic and market-related numerical data.

Figure 6 visually compares the forecasts of the three fusion strategies against the best baseline for weeks 1, 4, and 8. The plots confirm that the Early Fusion model tracks the actual price series most closely, especially during periods of high volatility.

4.3. Comparison with Recent Models

To rigorously position our work, we compare our best model (Early Fusion) with several recent SOTA crude oil price forecasting models in Table 6. We acknowledge that a direct, one-to-one comparison is challenging due to variations in datasets (WTI vs. Brent), forecast horizons (such as 1-Day vs. 4-Week), and evaluation periods. Nonetheless, this benchmark analysis provides a clear perspective on our model's performance relative to existing advanced methods.

Our model demonstrates a state-of-the-art combination of accuracy and multi-horizon capability. For a 4-week forecast, our Early Fusion model achieves an MAPE of 3.45% and an MSE of 6.81. This significantly outperforms the attention-PCA fusion model of Zhang et al. (2024) [23], which reported a MAPE of 4.11% and an MSE of 13.85 for a comparable 1-month horizon. It also surpasses the performance of the sentiment-enhanced model by Fang et

Table 4

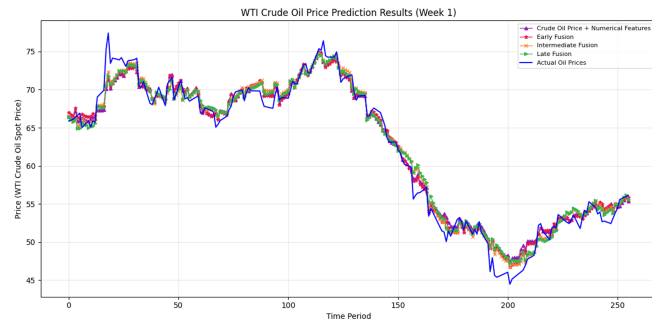
SMAPE-based performance comparison across modalities (%). Lower is better. Standard deviation in parentheses.

Metric (SMAPE %)	Crude Oil Price	Other Num. Features	Textual Features	Crude Oil + Textual	Crude Oil + Num.	Early Fusion	Intermediate Fusion	Late Fusion
Week 1	1.93 (0.34)	1.71 (0.21)	2.00 (0.45)	1.64 (0.24)	1.61 (0.27)	1.53 (0.20)	1.58 (0.17)	1.63 (0.14)
Week 2	3.13 (0.60)	2.26 (0.38)	3.25 (0.74)	2.58 (0.49)	2.09 (0.35)	1.92 (0.25)	2.06 (0.12)	2.21 (0.25)
Week 3	4.66 (0.82)	3.08 (0.57)	4.84 (0.93)	4.06 (0.80)	2.97 (0.50)	2.62 (0.45)	2.75 (0.31)	2.96 (0.44)
Week 4	5.78 (1.14)	3.78 (0.57)	5.94 (1.24)	5.27 (0.91)	3.60 (0.66)	3.40 (0.62)	3.52 (0.62)	3.75 (0.69)
Week 5	7.02 (1.49)	4.64 (0.82)	7.03 (1.69)	6.46 (1.22)	4.44 (0.90)	4.42 (0.78)	4.46 (1.00)	4.79 (0.97)
Week 6	7.79 (1.73)	5.12 (0.99)	7.59 (2.01)	7.11 (1.52)	4.88 (1.11)	4.84 (0.90)	4.89 (1.43)	5.28 (1.31)
Week 7	8.36 (1.98)	5.46 (1.33)	7.88 (2.30)	7.51 (1.80)	5.43 (1.49)	5.14 (1.19)	5.21 (1.70)	5.66 (1.57)
Week 8	9.05 (2.43)	6.15 (1.73)	8.32 (2.74)	7.90 (2.17)	6.10 (1.85)	5.51 (1.63)	5.74 (2.13)	6.20 (1.97)

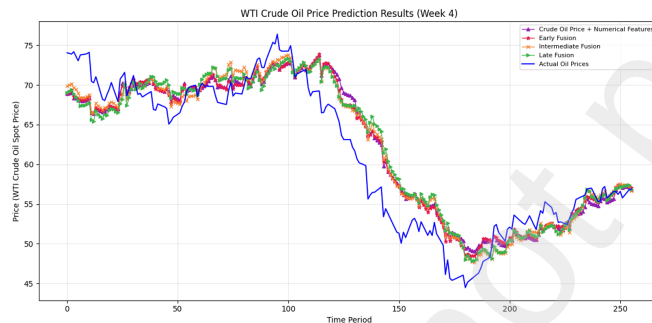
Table 5

Trend accuracy ($\pm 1\%$ threshold) across different modalities (%). Higher is better. Standard deviation in parentheses.

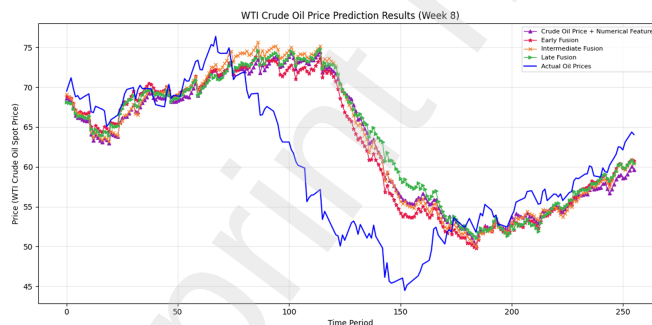
Trend Acc. (%)	Crude Oil Price	Other Num. Features	Textual Features	Crude Oil + Textual	Crude Oil + Num.	Early Fusion	Intermediate Fusion	Late Fusion
Week 1	66.4 (2.5)	69.1 (4.8)	64.6 (1.4)	71.4 (4.3)	72.0 (4.4)	70.2 (3.9)	71.7 (5.0)	71.4 (5.6)
Week 2	65.0 (8.6)	77.3 (5.9)	64.7 (6.1)	74.7 (6.2)	77.9 (6.7)	79.2 (4.8)	77.7 (6.4)	77.3 (4.7)
Week 3	59.2 (8.5)	74.9 (4.8)	55.7 (8.8)	67.1 (4.3)	76.7 (4.7)	81.5 (3.7)	78.3 (4.7)	78.3 (4.6)
Week 4	52.5 (8.1)	78.8 (0.8)	50.1 (8.3)	63.0 (2.7)	82.3 (1.6)	81.5 (3.8)	81.0 (3.6)	78.8 (5.1)
Week 5	49.2 (8.0)	71.7 (4.6)	44.7 (6.6)	58.1 (2.1)	76.2 (1.7)	73.4 (6.1)	78.1 (2.8)	71.0 (5.7)
Week 6	49.9 (10.4)	72.5 (4.5)	46.4 (5.7)	54.7 (2.5)	73.6 (3.3)	71.5 (4.7)	74.1 (2.1)	69.5 (4.7)
Week 7	47.5 (14.3)	68.8 (4.5)	48.1 (2.8)	54.0 (2.1)	67.6 (4.2)	69.8 (4.6)	70.3 (6.7)	66.9 (5.4)
Week 8	46.1 (16.4)	69.4 (2.9)	50.4 (2.1)	57.4 (2.6)	70.1 (6.6)	72.0 (6.2)	74.5 (6.4)	69.7 (4.3)



(a) Week 1 Forecasts



(b) Week 4 Forecasts



(c) Week 8 Forecasts

Figure 6: Comparison of WTI price forecasts from fusion strategies and the best baseline model for weeks 1, 4, and 8.

al. (2023) [11], which had a 1-week MAPE of 5.03%. While complex dual-decomposition models like Zhu et al. (2025) [30] are highly specialized for single-day-ahead predictions,

the primary strength of our framework lies in its **sustained accuracy over much longer periods**, maintaining a competitive MAPE of 5.78% even at an 8-week horizon.

4.4. Discussion

The superior performance of the early fusion strategy strongly suggests that enabling the model to learn complex, nonlinear interactions between numerical and textual features from a unified input vector is the most effective approach. By presenting all information jointly to the LSTM, the model can identify how macroeconomic trends are amplified, counteracted, or preempted by market sentiment and narratives expressed in financial news. The decomposition-based architecture further refines this process by allowing for the targeted application of features—for example, using our event-rich textual data to model the high-frequency residual component, thereby effectively capturing market shocks.

When benchmarked against the state-of-the-art models in Table 6, our framework's primary contribution becomes evident: its robust multi-horizon forecasting capability. The model's durable performance is a direct result of our novel feature engineering pipeline. The use of generative AI to transform raw news articles into high-quality, contextual summaries provides a probably persistent and potent predictive signal that does not decay as quickly as the signals used in many short-term models.

This ability to deliver accurate forecasts weeks in advance is of significant practical value, directly addressing the needs of practitioners who require reliable guidance for strategic decision-making in energy procurement [44], capital investment planning [45], and financial risk management. This distinguishes our framework from many existing models that, while accurate, excel primarily at short-term prediction and thus have limited utility for longer, more strategically relevant timeframes.

5. Conclusion

In this paper, we introduced a novel multimodal fusion framework for multi-horizon crude oil price forecasting that effectively integrates structured numerical data with generative AI-enhanced textual features. By decomposing the price

Table 6
Performance Comparison with Recent Crude Oil Price Forecasting Models.

Reference	Methodology	Horizon	MSE	MAPE (%)	SMAPE (%)	Trend Acc. (%)
This Study (Early Fusion)	STL-LSTM + GenAI Text	1-Week	1.37	1.54	1.53	70.2
This Study (Early Fusion)	STL-LSTM + GenAI Text	4-Week	6.81	3.45	3.40	81.5
This Study (Early Fusion)	STL-LSTM + GenAI Text	8-Week	23.59	5.78	5.51	72.0
Fang et al. (2023) [11]	FinBERT-VMD-Att-BiGRU	1-Week	4.31	5.03	4.75	70.2
Zhang et al. (2024) [23]	Attention-PCA + ABC Fusion	1-Month (4-Week)	13.85	4.11	-	74.29
Zhu et al. (2025) [30]	CEEMDAN-VMD + CNN-BiLSTM	1-Day	15.63	3.66	-	-

series and systematically evaluating different fusion strategies, we have demonstrated that a well-designed multimodal architecture can significantly improve predictive accuracy and provide robust decision support.

Our key findings are threefold. First, the use of generative AI to create context-aware summaries from news articles, combined with domain-specific FinBERT embeddings, provides a powerful way to represent unstructured textual information. Second, a decomposition-based approach that models the trend, seasonal, and residual components separately with tailored inputs enhances both model performance and interpretability. Third, among the fusion strategies, early fusion consistently delivered the most accurate and stable forecasts, highlighting the benefit of learning joint representations from heterogeneous data at the earliest stage.

This research contributes a practical and powerful forecasting system for a highly complex market. The framework is not only accurate but also adaptive, capable of incorporating event-driven news signals. The multi-horizon forecasts are directly applicable to the long-term planning needs of energy procurement managers and financial analysts.

Future work could explore more advanced transformer-based architectures, such as the Temporal Fusion Transformer (TFT), which may offer further improvements in performance and interpretability. Additionally, incorporating topic modeling to select a more diverse set of daily news articles could reduce redundancy and enrich the textual input. Finally, exploring even larger language models for feature extraction could unlock deeper semantic insights from financial texts.

Additional Forecasting Results

This appendix provides additional visualizations of the forecasting results. Figures 7 through 11 show the performance of the best baseline model against the three fusion strategies for weeks 2, 3, 5, 6, and 7. Each figure presents a comparison between actual crude oil prices and forecasts generated by four distinct model configurations, each embodying a unique multimodal integration strategy. These configurations include:

1. **Baseline model:** The best-performing reference model using only structured numerical inputs, such as crude oil prices and macroeconomic indicators.

2. **Early fusion model:** Integrates numerical and textual features at the input level, enabling joint representation learning from the outset.
3. **Intermediate fusion model:** Processes each modality independently before merging their latent representations, allowing modality-specific feature extraction.
4. **Late fusion model:** Combines the final outputs of separate modality-specific models, emphasizing decision-level integration.

These visualizations complement the quantitative analysis in Section 4.2 by offering a qualitative perspective on how different fusion strategies affect forecasting accuracy. Notably, the early fusion model demonstrates the most consistent alignment with actual price trends across multiple weeks. This suggests that integrating structured data with news-derived features at the input stage enhances the model's ability to capture both economic signals and real-world events, leading to more robust predictions.

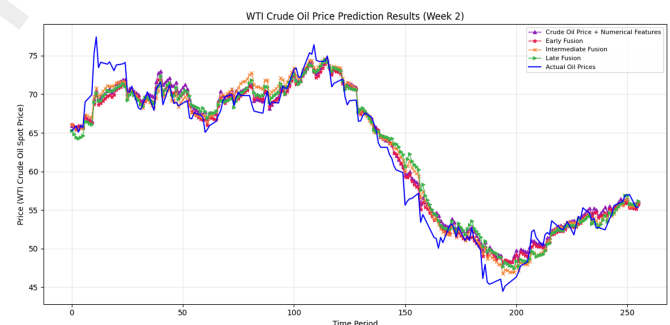


Figure 7: Forecasts for Week 2: Best baseline vs. fusion strategies.

Forecasting Results Across All Baseline and Fusion Strategies

Figure 12 visualizes the weekly WTI crude oil spot price forecasting results for the full eight-week horizon. Each plot compares the actual crude oil prices with predictions generated by all eight modality combinations, which include five baseline models and the three fusion strategies.

The baseline models are: (1) price-only, using historical crude oil prices; (2) numerical-only, using other structured economic features; (3) textual-only, using news-based inputs; (4) a combination of price and textual features; and

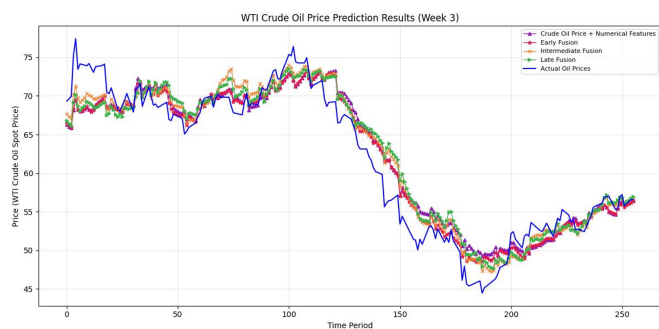


Figure 8: Forecasts for Week 3: Best baseline vs. fusion strategies.

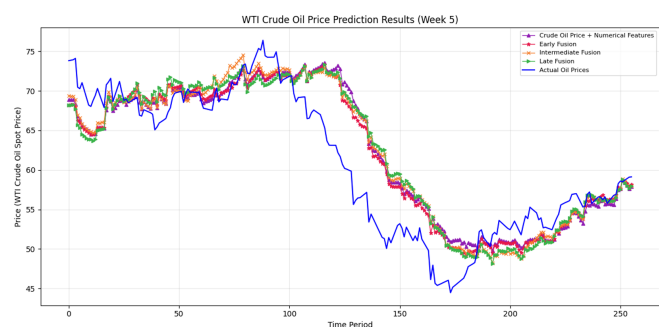


Figure 9: Forecasts for Week 5: Best baseline vs. fusion strategies.

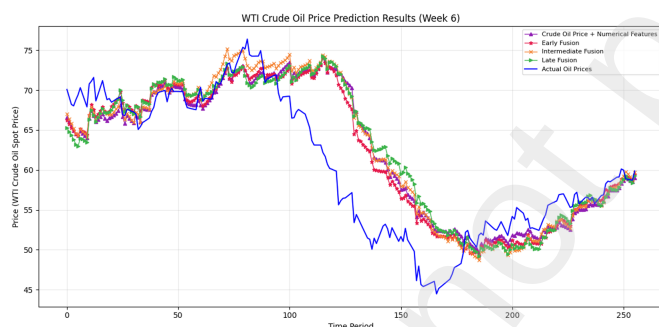


Figure 10: Forecasts for Week 6: Best baseline vs. fusion strategies.

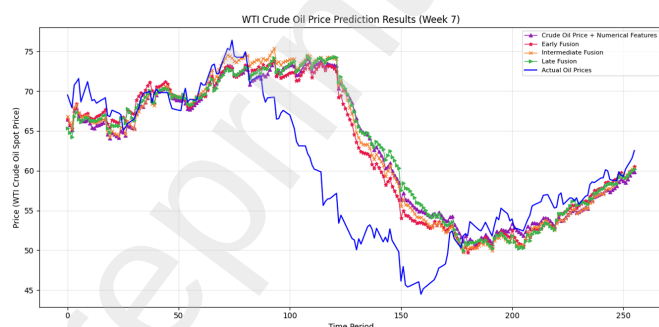


Figure 11: Forecasts for Week 7: Best baseline vs. fusion strategies.

(5) a combination of price and other numerical features.

The three fusion strategies are early, intermediate, and late fusion.

These visualizations extend the qualitative analysis by providing a broader perspective on model behavior across the entire forecasting horizon. The results reinforce the strength of fusion-based strategies, particularly early fusion, as well as powerful modality pairings such as price combined with numerical features. In contrast, single-modality baselines (e.g., price-only and textual-only) tend to perform less consistently, especially during volatile periods.

Detailed for Text Feature Generation

To address the limitations of short headlines, this study uses GPT-4.1 to generate enhanced textual features from the full content of commodity-related news articles. The input excludes the original headlines to ensure that the generated outputs are entirely based on the article body and remain non-redundant. Short headlines often prioritize attention-grabbing over clarity, using subjective or sensational language that may lack the contextual depth needed for reliable modeling. To mitigate this, the model adopts the role of a “helpful economist and journalist” and responds to each article using a structured set of prompts specifically designed to produce meaningful features. This study generates three types of additional textual features for each news article:

This study configures the model with a temperature setting of 0 to reduce randomness and improve consistency across generations. However, due to the inherent stochasticity of generative language models, this setting does not guarantee complete determinism across runs. Despite this, the outputs remain reliably structured and semantically aligned with their intended feature types.

Analysis of GPT-Generated Textual Features

The news articles utilized in this research primarily focus on developments within commodity markets, with an emphasis on crude oil, gold, silver, copper, and natural gas. These articles typically report on recent price movements and analyze the underlying causes, such as supply-demand imbalances, geopolitical events, and shifts in investor sentiment. They also often highlight upcoming events that could influence future prices, including inventory reports and central bank decisions. This combination of retrospective analysis and forward-looking signals provides a rich context for understanding crude oil price dynamics.

To transform this unstructured news into model-ready inputs, this study uses GPT to generate three types of textual features: rewritten headlines, keyword lists, and concise summaries. As illustrated in Table 7, the generated features offer different levels of context and clarity.

While generated headlines and keywords provide relevant signals, they capture only a limited portion of the market narrative. In contrast, the summaries offer a more comprehensive view by linking price changes to their causes and incorporating forward-looking information. For this reason, the summaries are selected as the primary textual input for

Table 7
Examples of GPT-generated features from commodity news articles.

Original Headline	Generated Headline	Extracted Keywords	Generated Summary
Oil Futures Trim Losses, WTI Rises Back Above \$45 in Volatile Trade	Oil Prices Rebound Above \$45 Amid Volatile Trading and Supply Concerns	Oil Prices, WTI, Brent, Goldman Sachs, Supply Glut.	Oil futures rebounded above \$45 a barrel after sharp declines, as concerns over global oversupply persisted and Goldman Sachs slashed its 2015 price forecasts for both WTI and Brent crude. Traders are awaiting new U.S. inventory data and market reports for further direction.
Crude Oil Futures Decline as Investors Weigh Supply Outlook	Oil Prices Fall Again as Global Supply Glut Persists	Oil Prices, Global Supply, Brent Crude, WTI, OPEC.	Oil futures declined on Monday amid ongoing concerns about oversupply, with both Brent and WTI prices falling despite recent gains driven by reduced non-OPEC output forecasts and a drop in U.S. drilling rigs. Market activity was subdued due to a U.S. holiday, while currency markets remained volatile ahead of expected ECB action.
Gold Declines as Investors Digest Greece Election Results	Gold, Silver, and Copper Prices Fall as Greek Election Spurs Market Volatility	Gold, Greece, Syriza, Commodities, Euro.	Gold and silver prices declined as markets reacted to Greece's anti-austerity Syriza party winning the election, raising concerns over eurozone stability and impacting global commodities. Investors are also watching the upcoming Federal Reserve meeting and U.S. economic data for further direction.

the forecasting model, as they enhance the quality of features and bridge the gap between natural language and machine-readable input.

Price Decomposition and Stationarity Analysis

To improve interpretability and modeling precision, this study employs STL to decompose the WTI crude oil spot price series into three additive components: trend, seasonality, and residual. As shown in Figure 13, this decomposition isolates long-term movements (trend), recurring industry cycles (seasonal), and irregular price swings (residual). This structure strengthens the forecasting framework by allowing the model to align each component with the most relevant inputs.

The Augmented Dickey-Fuller (ADF) test was applied to the WTI price and its decomposed components to ensure valid time-series modeling. The results in Table 8 show that the original WTI price and the trend component are non-stationary but become stationary after first-order differencing. The seasonal and residual components are inherently stationary. To avoid distorting the data, differencing is applied only to the trend component, while the seasonal and residual parts are retained in their original form.

Acknowledgement

This work was supported by the National Science and Technology Council (NSTC), Taiwan R.O.C., under Grant NSTC 114-2221-E002-217, NSTC 112-2634-F-011-002-MBK, Grant NSTC 113-2634-F-011-002-MBK, and Grant

NSTC 111-2221-E-A49-202-MY3. Further support was provided by the Green Energy & Environment Research Laboratories, Industrial Technology Research Institute, Taiwan, R.O.C. Additionally, this work received partial funding from the National Taiwan University (NTU) under Grant NTU-JP-113L7256, Grant NTU-CC-114L895501, and Grant NTU-G0647 within the Higher Education Sprout Project framework by the Ministry of Education, Taiwan. Thanks to Chung-Wei Chuang at National Taiwan University for implementing part of the software. Thanks to Chung-Wei Chuang for writing the partial software. Thanks to Jui-Ning Chen at Google and Academia Sinica for her valuable comments. Guan-Yan Yang is grateful to the National Science and Technology Council (NSTC) in Taiwan for the graduate research fellowship (NSTC-GRF) and to Professor Hung-Yi Lee for co-hosting his Ph.D research project.

CRedit authorship contribution statement

Guan-Yan Yang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Resource, Supervision. **Wei-Yang Chiu:** Conceptualization, Investigation, Writing - Review & Editing. **Farn Wang:** Validation, Supervision, Project administration. **Kuo-Hui Yeh:** Resource, Writing - Review & Editing, Supervision, Project administration.

References

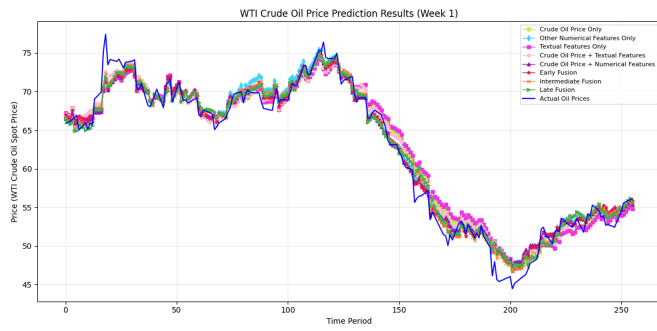
- [1] Ankit Kumar Patidar, Prateek Jain, Prateek Dhasmana, and Tota Choudhury. Impact of global events on crude oil economy: a

Table 8

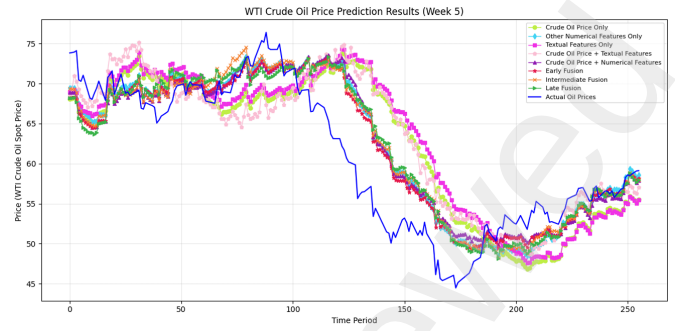
ADF test results for WTI price and decomposed components.

Target Variable (y)	No Differencing (p-value)	Differencing Once (p-value)
WTI Crude Oil Spot Price	0.3001 (non-stationary)	<0.001 (stationary)
Trend Component	0.1118 (non-stationary)	<0.001 (stationary)
Seasonal Component	<0.001 (stationary)	-
Residual Component	<0.001 (stationary)	-

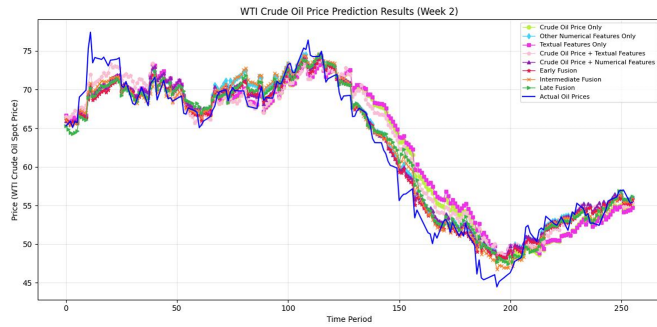
- comprehensive review of the geopolitics of energy and economic polarization. *GeoJournal*, 89(2):56, 2024.
- [2] Refk Selmi, Shawkat Hammoudeh, and Mark E Wohar. What drives most jumps in global crude oil prices? fundamental shortage conditions, cartel, geopolitics or the behavior of financial market participants. *The World Economy*, 46(3):598–618, 2023.
- [3] Muhammad Bilal, Muhammad Aamir, Sadiq Abdullah, and Faheem Khan. Impacts of crude oil market on global economy: Evidence from the ukraine-russia conflict via fuzzy models. *Heliyon*, 10(3), 2024.
- [4] Mengxuan Li. Prediction of the west texas intermediate crude oil price using arima model and arima-garch model. *Proceedings of the 1st International Conference on Innovations in Applied Mathematics, Physics and Astronomy*, 1:110–118, 2024.
- [5] Ying Xiang. Using arima-garch model to analyze fluctuation law of international oil price. *Mathematical Problems in Engineering*, 2022(1):3936414, 2022.
- [6] Lutz Kilian and Xiaoqing Zhou. The econometrics of oil market var models. Technical report, Federal Reserve Bank of Dallas, 2020.
- [7] Zhaosheng Cui, Tianen Li, Zhong Ding, Xiaojun Li, and Jianshe Wu. Probabilistic oil price forecasting with a variational mode decomposition-gated recurrent unit model incorporating pinball loss. *Data Science and Management*, 2024.
- [8] Guohua Zheng, Yanjun Li, and Yuchen Xia. Crude oil price forecasting model based on neural networks and error correction. *Applied Sciences*, 15(3):1055, 2025.
- [9] Xiao Li, Wenting Shang, and Shouyang Wang. Text-based crude oil price forecasting: A deep learning approach. *International Journal of Forecasting*, 35(4):1548–1560, 2019.
- [10] Yang Bai, Xiao Li, Hong Yu, and Shaohui Jia. Crude oil price forecasting incorporating news text. *International Journal of Forecasting*, 38(1):367–383, 2022.
- [11] Yue Fang, Weimin Wang, Peng Wu, and Yanchu Zhao. A sentiment-enhanced hybrid model for crude oil price forecasting. *Expert Systems with Applications*, 215:119329, 2023.
- [12] Tian Shi, Kyeong-Myeong Kang, Jaegul Choo, and Chandan K Reddy. Short-text topic modeling via non-negative matrix factorization enriched with local word-context correlations. In *The Web Conference 2018*, pages 1105–1114, 2018.
- [13] Yang Cao, Yangsong Lan, Feiyan Zhai, and Piji Li. Sw1h extraction with large language models. In *2024 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8, 2024.
- [14] Pin-Yu Chen, Zois Boukouvalas, and Roberto Corizzo. A deep fusion model for stock market prediction with news headlines and time series data. *Neural Computing and Applications*, 36(24):21229–21271, 2024.
- [15] L Gasper and H Mbwambo. Forecasting crude oil prices by using arima model: Evidence from tanzania. *Journal of Accounting Finance and Auditing Studies (JAFAS)*, 9(2):1–15, 2023.
- [16] Yue-Jun Zhang and Han Zhang. Volatility forecasting of crude oil market: which structural change based garch models have better performance? *The Energy Journal*, 44(1):175–194, 2023.
- [17] Xia-Jane He. Crude oil prices forecasting: time series vs. svr models. *Journal of International Technology and Information Management*, 27(2):2, 2018.
- [18] Halah Aldabagh, Xinjue Zheng, Mojtaba Najand, and Raghunath Mukkamala. Forecasting crude oil price using multiple factors. *Journal of Risk and Financial Management*, 17(9):415, 2024.
- [19] Haruna Chiroma, Sameem Abdulkareem, and Tutut Herawan. Evolutionary neural network model for west texas intermediate crude oil price prediction. *Applied Energy*, 142:266–273, 2015.
- [20] Amir Daneshvar, Maryam Ebrahimi, Fariba Salahi, Maryam Rahmaty, and Mahdi Homayounfar. Brent crude oil price forecast utilizing deep neural network architectures. *Computational Intelligence and Neuroscience*, 2022(1):6140796, 2022.
- [21] Kai Zhang and Ming Hong. Forecasting crude oil price using lstm neural networks. *Data Science in Finance and Economics*, 2(3):163–180, 2022.
- [22] Bo Jin and Xiaohua Xu. Price forecasting through neural networks for crude oil, heating oil, and natural gas. *Measurement: Energy*, 1:100001, 2024.
- [23] Xiao Zhang, Sheng Cheng, Yifei Zhang, Jue Wang, and Shouyang Wang. An attention-pca based forecast combination approach to crude oil price. *Expert Systems with Applications*, 240:122463, 2024.
- [24] Lean Yu, Zishu Wang, and Ling Tang. A decomposition-ensemble model with data-characteristic-driven reconstruction for crude oil price forecasting. *Applied Energy*, 156:251–267, 2015.
- [25] Yujie Lin, Kaifang Chen, Xinyu Zhang, Bowen Tan, and Qing Lu. Forecasting crude oil futures prices using bilstm-attention-cnn model with wavelet transform. *Applied Soft Computing*, 130:109723, 2022.
- [26] Ani Shabri and Ruhaidah Samsudin. Daily crude oil price forecasting using hybridizing wavelet and artificial neural network model. *Mathematical Problems in Engineering*, 2014(1):201402, 2014.
- [27] Tianxiang Fang, Chuqiao Zheng, and Dong Wang. Forecasting the crude oil prices with an emd-isbm-fnn model. *Energy*, 263:125407, 2023.
- [28] Hooman Abdollahi. A novel hybrid model for forecasting crude oil price based on time series decomposition. *Applied Energy*, 267:115035, 2020.
- [29] Hao Gong, Haiyang Xing, Yuanyuan Yu, and Yanhui Liang. A combined model using secondary decomposition for crude oil futures price and volatility forecasting: Analysis based on comparison and ablation experiments. *Expert Systems with Applications*, 252:124196, 2024.
- [30] Shijie Zhu, Mei Xu, Jie Wu, Yaning Wang, Xinsheng Jiang, Zhuangzhuang Huang, Yang Wang, and Yong Zhu. A study on crude oil price forecasting model integrating ceemdan-vmd multiscale decomposition with cnn-bilstm. *Results in Engineering*, 27:106391, 2025.
- [31] Yelin Li, Shengguo Jiang, Xiao Li, and Shouyang Wang. The role of news sentiment in oil futures returns and volatility forecasting: Data-decomposition based deep learning approach. *Energy Economics*, 95:105140, 2021.
- [32] Lei Ding, Hong Zhao, and Ruijie Zhang. Predicting multi-frequency crude oil price dynamics: Based on midas and stl methods. *Energy*, 313:134003, 2024.
- [33] Ling Tang, Chengyuan Zhang, Ling Li, and Shouyang Wang. A multi-scale method for forecasting oil price with multi-factor search engine data. *Applied Energy*, 257:114033, 2020.
- [34] Bing Wu, Lin Wang, Shao-Xiong Lv, and Yan-Ru Zeng. Effective crude oil price forecasting using new text-based and big-data-driven model. *Measurement*, 168:108468, 2021.
- [35] Ziyuan Liu, Deyi Huang, Kangle Huang, Zhaocheng Li, and Jun Zhao. Finbert: A pre-trained financial language representation model for financial text mining. In *Proceedings of the 28th International*



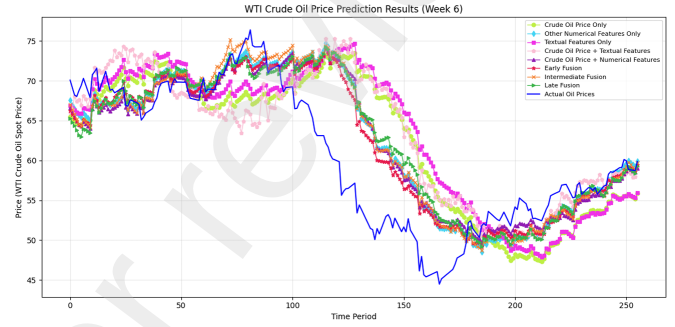
(a) Forecasting results for Week 1: All baseline models vs. three fusion strategies.



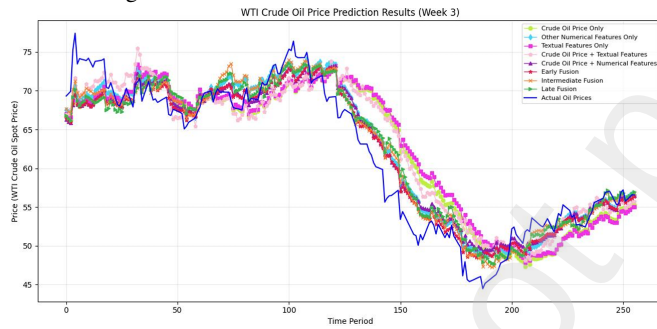
(e) Forecasting results for Week 5: All baseline models vs. three fusion strategies.



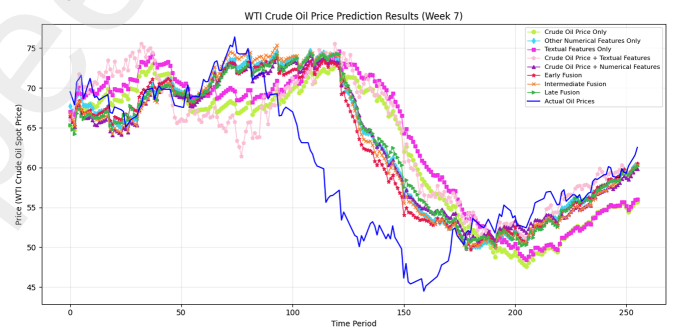
(b) Forecasting results for Week 2: All baseline models vs. three fusion strategies.



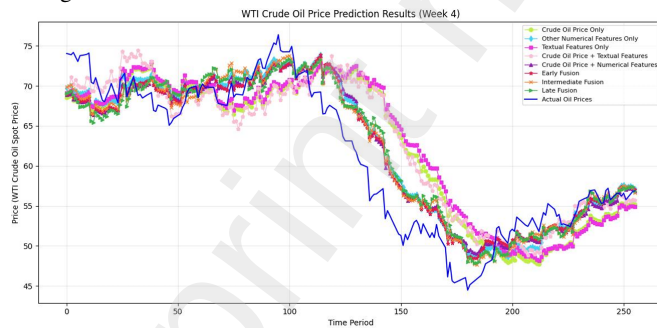
(f) Forecasting results for Week 6: All baseline models vs. three fusion strategies.



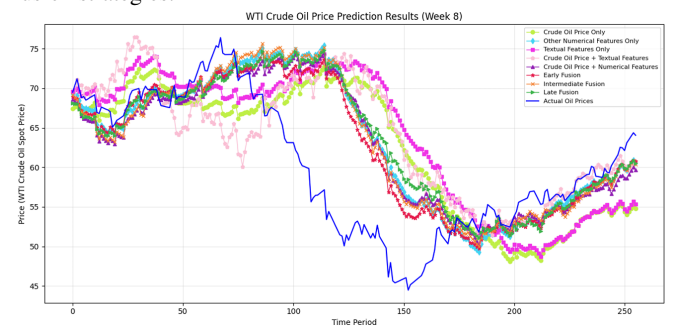
(c) Forecasting results for Week 3: All baseline models vs. three fusion strategies.



(g) Forecasting results for Week 7: All baseline models vs. three fusion strategies.



(d) Forecasting results for Week 4: All baseline models vs. three fusion strategies.



(h) Forecasting results for Week 8: All baseline models vs. three fusion strategies.

Figure 12: Weeks 1–4: All baselines vs. fusion strategies.

Figure 12: Weeks 5–8: All baselines vs. fusion strategies.

Conference on Computational Linguistics, pages 6433–6444, 2020.

- [36] L Blair Anderson, Deekshitha Kanneganti, Michael B Houk, Rochelle H Holm, and Tami Smith. Generative ai as a tool for environmental health research translation. *GeoHealth*,

7(7):e2023GH000867, 2023.

- [37] Yancheng Jiang, Kaixuan Ning, Zhaofeng Pan, Xiang Shen, Jiaming Ni, Wei Yu, Andrew Schneider, Haofen Chen, Yury Nevmyvaka, and Dawn Song. Multi-modal time series analysis: A tutorial and survey.

Generated Headline

A concise, high-quality headline that is attractive, clear, accurate, and inoffensive.

Prompt: "Please come up with a high-quality (attractive, clear, accurate, and inoffensive) headline for the following news article; please only generate the headline without quotation marks."

Keywords

Five short keywords that capture the main themes of the article.

Prompt: "Please identify five keywords for the following news article, and make each one or two words long (when generating the keywords, please use the format: ..., ..., ...)."

Summary

A one-line summary that conveys the article's key message as concisely as possible.

Prompt: "Generate a short summary of the following news article in as few words as possible; please ensure the summary is in one line without creating a new line for a new sentence."

arXiv preprint arXiv:2501.01859, 2025.

- [38] Yuxin Pei, John Cartlidge, Arijit Mandal, David Gold, Wladimir Marcilio, and Raphael Mazzon. Cross-modal temporal fusion for financial market forecasting. *arXiv preprint arXiv:2501.03606*, 2025.
- [39] Haoyang Wang, Zuopeng Xie, Dickson KW Chiu, and Kevin KW Ho. Multimodal market information fusion for stock price trend prediction in the pharmaceutical sector: Multimodal market information fusion for stock price trend prediction in the pharmaceutical sector. *Applied Intelligence*, 55(1):1120–1143, 2025.
- [40] Edward Windsor and Wen Cao. Improving exchange rate forecasting via a new deep multimodal fusion model. *Applied Intelligence*, 52(14):16701–16717, 2022.
- [41] Soufiane Akil, Soukaina Sekkate, and Abdellah Adib. Multimodal deep learning for oil price forecasting using economic indicators. *Procedia Computer Science*, 236:402–409, 2024.
- [42] Peng Han, Haodong Chen, Abid Rasool, Qing Jiang, and Meng Yang. Mfb: A generalized multimodal fusion approach for bitcoin price prediction using time-lagged sentiment and indicator features. *Expert Systems with Applications*, 261:125515, 2025.
- [43] Hendrik Kaplan, Ralf-Peter Mundani, Heiko Rölke, and Albert Wechselbraun. Crudebert: Applying economic theory towards fine-tuning transformer-based sentiment analysis models to the crude oil market. *arXiv preprint arXiv:2305.10174*, 2023.
- [44] Rui Chen, Tiemeng Deng, Simin Huang, and Ran Qin. Optimal crude oil procurement under fluctuating price in an oil refinery. *European Journal of Operational Research*, 245(2):438–445, 2015.
- [45] Todd R Zabelle and Matthew Parsons. Consequence of lead time on capital project outcomes. *Journal of Project Production Management*, 2:25–34, 2017.

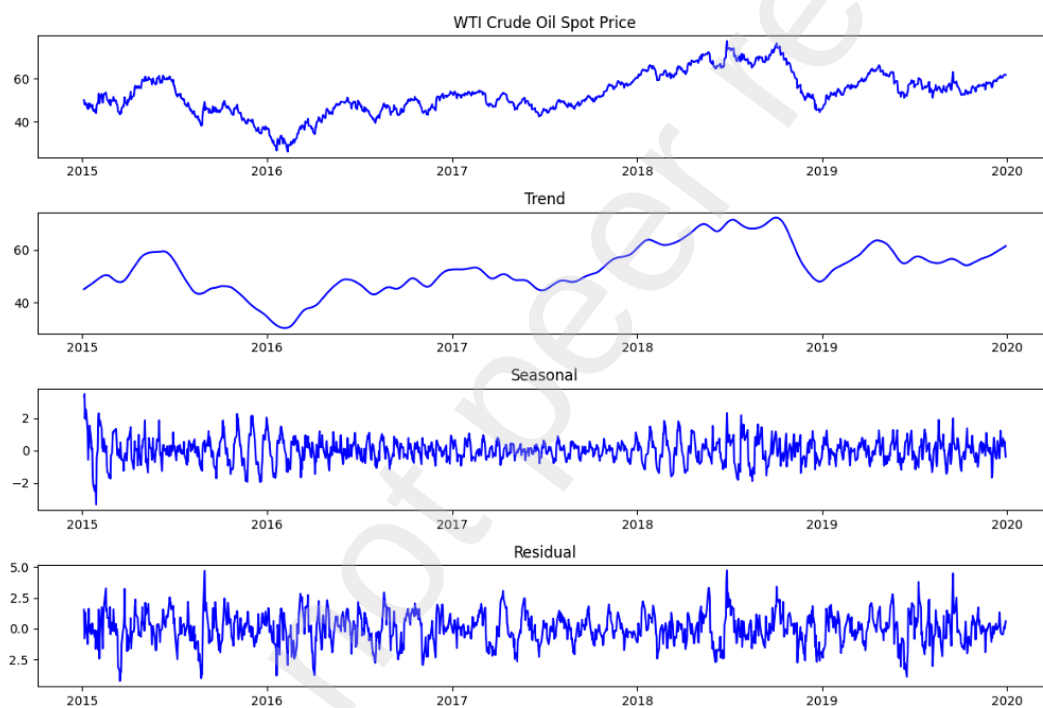


Figure 13: Graphical representation of WTI price and its decomposed components.